Chreode: A Cell World Model for One-Step Temporal Dynamics and Perturbation Prediction

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Abstract

Predicting how a cell will change its transcriptional state under a developmental signal or a genetic perturbation is the computational core of in-silico biology and the AI Virtual Cell program. Existing approaches either fit static control-to-treated maps that discard time, or solve multi-step ODE / Schrödinger-bridge problems on each dataset independently. We introduce **Chreode**, a one-step cell world model that predicts action-conditioned cell-state transitions through a structured residual transition operator. It shifts distributional evolution from inference time to training time, enabling single-pass generation while preserving a Waddington-inspired decomposition into downhill landscape flow, rotational in-tangent dynamics, and stochastic spread. The model is pretrained with a shared scVI encoder and a DiT-based dynamics backbone on a 2.4M-cell mouse embryonic atlas spanning 7 datasets. As a fine-tuning initialization, **Chreode** improves per-target Sinkhorn distance on Weinreb hematopoiesis and Veres islet differentiation over matched scratch models, PI-SDE, and PRESCIENT. As a transferable gene-state embedding for GEARS, the pretrained dynamics representation reduces shared-vocabulary DE20 mean squared error on Norman Perturb-seq from 0.2121 to 0.1858, a 12.4% relative improvement, without changing the GEARS training procedure. We interpret this transfer to perturbation prediction as evidence that pretrained developmental-trajectory dynamics encode differentiation primitives transferable to CRISPR-induced state shifts, since both involve cell-state transitions in a shared latent geometry. The pretrained backbone additionally produces zero-shot clonal fate scores on Weinreb that are competitive with strong dynamic-OT baselines.

1 Introduction

A modern in-silico perturbation screen needs to query on the order of 10^4 cells, 10^3 drugs or genetic perturbations, and 10 time points, of order 10^8 (state, action, time) tuples per screen. Existing generative cell-dynamics models cost 40 to 100 ODE or stochastic-bridge steps per query, putting full screens at 10^{10} network forward passes. Predicting how a cell will change its transcriptional state given an action and an elapsed time is the computational core of the *AI Virtual Cell* program [Bunne et al., 2024]; reaching screen scale requires amortizing the transition itself, not just running the same multi-step solver faster.

We call a model that directly predicts this transition a *cell world model*, by analogy to the state-action-transition world models in reinforcement learning [Ha and Schmidhuber, 2018, Hafner et al., 2020]. Given a latent transcriptional state z_t at time t, an intervention a, and an elapsed time Δ, such a model predicts the distribution of possible future states $z_{t+\Delta}$, equivalently $p(z_{t+\Delta} \mid z_t, \mathrm{do}(a))$. We use the analogy as a predictive-model framing only and do not claim planning or rollout.

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Four properties of cell dynamics make a naive application of existing generative recipes fail. (i) *Temporal information is systematically underused*: destructive scRNA-seq yields unpaired population snapshots, yet most perturbation predictors collapse these into static control-to-treated maps and discard the rate of change biology provides for free [Lotfollahi et al., 2019, 2023, Roohani et al., 2024, Bunne et al., 2023]. (ii) *State and action are not interchangeable*: a drug is an *intervention* in the sense of Pearl's do-calculus [Pearl, 2009], not a latent context to concatenate; simple additive composition [Lotfollahi et al., 2019, 2023] is accurate near attractors but cannot resolve compositional or trajectory-dependent perturbations. (iii) *Inference cost does not amortize with screen scale*: per-query multi-step integration multiplies linearly with the number of (cell, action, time) tuples, so a 10^2-step solver makes a 10^8-query screen cost 10^{10} forward passes regardless of per-step efficiency. (iv) *The inductive bias is biological rather than Euclidean*: Waddington's landscape [Waddington, 1957], cell fate as a ball rolling through a potential with rotational currents, motivates a residual decomposed into a potential gradient, an antisymmetric flow, and a stochastic spread; yet existing flow-matching, Schrödinger-bridge, and OT methods [Lipman et al., 2023, Tong et al., 2023, Tang et al., 2025, Klein et al., 2025a] parameterize the velocity field as a generic neural network without this architectural reflection.

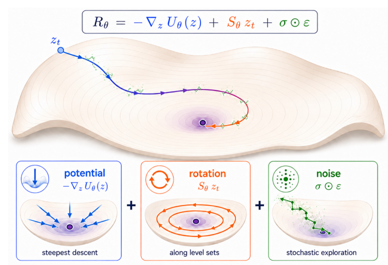

Figure 1: The Waddington residual at the heart of Chreode. A latent state z_t traverses a learned landscape via a potential gradient $-\nabla_z U_\theta$ (blue), an antisymmetric in-tangent rotation $S_\theta z_t$ (orange), and stochastic spread $\sigma \odot \epsilon$ (green). The full residual is applied as a single one-step update $\hat{z}_{t+\Delta} = z_t + \alpha(\Delta) R_\theta$, not by ODE rollout.

Prior work addresses at most two of these properties. Flow-matching and Schrödinger-bridge methods [Lipman et al., 2023, Tong et al., 2023, Tang et al., 2025, Klein et al., 2025a] use time, but require multi-step integration and typically lack explicit action conditioning. Perturbation predictors such as scGen, CPA, GEARS, and CellOT [Lotfollahi et al., 2019, 2023, Roohani et al., 2024, Bunne et al., 2023] model actions, but fit static control-to-treated maps and ignore temporal structure. Cell foundation models such as scGPT, Geneformer, and CellStream [Cui et al., 2024, Theodoris et al., 2023, Ling et al., 2025] learn transferable representations, but do not natively model a transition operator. Concurrent cellular world models, AlphaCell [Chuai et al., 2026] and X-Cell [Wang et al., 2026], scale the framing, but rely on multi-step OT-CFM or iterative diffusion denoising. We instead build on the recent *Drifting Models* framework [Deng et al., 2026], which shows that a single-forward-pass network can match multi-step diffusion / flow methods by evolving the pushforward distribution *at training time* via a population-level drifting field; we combine this training-time pushforward with a Waddington biological prior and snapshot-atlas pretraining.

Our approach. We introduce **Chreode**[1]. For an input latent state z_t, elapsed time Δ, intervention a, and random noise vector ϵ, Chreode predicts a future latent state $\hat{z}_{t+\Delta}$ by adding a learned residual to the input state:

$$\hat{z}_{t+\Delta} \;=\; z_t \;+\; \alpha(\Delta) \cdot R_\theta(z_t, \Delta, a, \epsilon), \qquad R_\theta \;=\; -\nabla_z U_\theta \;+\; S_\theta\, z_t \;+\; \sigma_\theta \odot \epsilon, \tag{1}$$

Here, R_θ is the learned residual transition, and θ is the trainable neural network parameters. The first term, $-\nabla_z U_\theta$, is the negative gradient of a learned potential function U_θ with respect to the latent state z_t; it represents downhill motion on a Waddington-style landscape. The second term, $S_\theta z_t$, uses a learned antisymmetric operator S_θ to model rotational or cyclic dynamics. In practice, we parameterize this operator as $S_\theta = P_\theta Q_\theta^\top - Q_\theta P_\theta^\top$, where P_θ and Q_θ are neural network outputs whose difference of outer products guarantees antisymmetry. The third term, $\sigma_\theta \odot \epsilon$, models stochastic population spread, where σ_θ is a learned scale vector, ϵ is sampled noise, and $\odot$ denotes elementwise multiplication. Finally, $\alpha(\Delta) = 1 - e^{-\Delta/\tau_0}$ is a time gate with learnable time constant τ_0; this gate ensures that the predicted change vanishes when no time has elapsed, because $\alpha(0) = 0$. We pretrain Chreode in two stages: a shared scVI [Lopez et al., 2018] encoder over a fixed cross-species ortholog vocabulary, then a DiT-based dynamics backbone [Peebles and Xie, 2023] trained on a 2.4M-cell mouse embryonic atlas of seven datasets and 88 timepoints, using a population-level drifting-field loss [Deng et al., 2026]. This loss provides anti-collapse gradients beyond maximum mean discrepancy, denoted MMD, and Sinkhorn Wasserstein-2 distance, denoted W_2.

[1]A *chreode* [Waddington, 1957] is a canalized developmental trajectory through the epigenetic landscape. We use this term because Chreode predicts a single-step transition along a learned canalized flow.

After pretraining, the frozen Chreode backbone provides two distinct transfer modes. As a *fine-tuning initialization* for population time-transition prediction (§5.1, §5.2), the pretrained backbone improves per-target Sinkhorn W_2 on Weinreb hematopoiesis ($d4$, $d6$) and Veres islet differentiation ($t1$ through $t7$) over matched scratch and over PI-SDE / PRESCIENT [Yeo et al., 2021] trained on the same shared scVI-128 latent. As a *gene-state embedding* injected into the existing GEARS predictor [Roohani et al., 2024] on Norman Perturb-seq (§5.4), it reduces shared-vocabulary DE20 MSE from 0.2121 to 0.1858 (12.4% relative) without modifying the GEARS training procedure; we interpret this as evidence that the dynamics backbone, pretrained on mouse embryonic differentiation manifolds, encodes primitives that also describe CRISPR-induced state shifts. The pretrained backbone additionally produces zero-shot clonal fate scores on Weinreb (§5.3) competitive with strong dynamic-OT baselines (moscot, WOT, scDiffEq).

Contributions.

- **Pretraining recipe** (§4): a two-stage pipeline of a shared scVI encoder over a $16{,}520$-gene cross-species ortholog vocabulary and a one-step Waddington-DiT dynamics backbone, trained on a 2.4M-cell mouse embryonic atlas spanning seven datasets and 88 timepoints, without any optimal-transport supervision.
- **Architecture** (§3): a one-step action-conditioned residual with a Waddington-style decomposition into a potential gradient, an antisymmetric flow component, and a stochastic spread, with decoupled time embeddings for the gradient and rotational branches.
- **Time-transition transfer** (§5.1, §5.2, §5.3): used as fine-tuning initialization, the pretrained backbone improves per-target Sinkhorn W_2 on Weinreb $d4/d6$ and on every Veres target $t1$ through $t7$ over matched scratch, PI-SDE, and PRESCIENT, and produces zero-shot clonal fate scores competitive with strong dynamic-OT baselines.
- **Perturbation embedding transfer** (§5.4): injected as the gene-state embedding inside GEARS, the pretrained dynamics representation reduces shared-vocabulary DE20 MSE on Norman Perturb-seq by 12.4%, evidence that pretrained developmental-trajectory dynamics encode differentiation primitives transferable to CRISPR-induced perturbation prediction.

Scope. This paper is about a controlled latent transition operator with a structured biological prior. We do not claim a general-purpose cell representation (Theodoris et al. 2023, Cui et al. 2024 are stronger on that axis), we do not propose a knowledge-rich gene perturbation encoder (Chuai et al. 2026, Wang et al. 2026 go further on that axis), and we do not benchmark against closed-source 4.9B-parameter systems whose pretraining data is proprietary. Our contribution is the structured one-step generator and its pretraining recipe; we expect the perturbation encoder to evolve in follow-up work without changing the residual backbone.

2 Related work

Static perturbation predictors. A first family of methods treats perturbation response as a static map from a control population to a treated population. scGen adds a learned latent shift [Lotfollahi et al., 2019]; CPA disentangles cellular state from a compositional perturbation embedding [Lotfollahi et al., 2023]; GEARS couples a gene-graph prior with a graph neural network to predict multi-gene knockouts [Roohani et al., 2024]; and CellOT learns a neural optimal-transport map between control and perturbed marginals [Bunne et al., 2023]. These methods model actions well, but discard temporal information: they fit one response per perturbation and do not answer queries at arbitrary time horizons. Chreode instead models the controlled transition $p(z_{t+\Delta} \mid z_t, \mathrm{do}(a))$ so that a trajectory at any Δ and any action is a single forward pass from the same backbone.

Dynamics from snapshot populations. Because scRNA-seq is destructive, cell dynamics are often inferred from unpaired snapshots. Waddington-OT estimates stochastic couplings between time points [Schiebinger et al., 2019]; TrajectoryNet links continuous normalizing flows with dynamic optimal transport [Tong et al., 2020]; PRESCIENT learns a stochastic potential landscape that can be integrated forward under interventions [Yeo et al., 2021]; and dynamo reconstructs transcriptomic vector fields from velocity information [Qiu et al., 2022]. Recent flow-matching and Schrödinger-bridge variants further improve generative population transport: CFM with minibatch OT [Lipman et al., 2023, Tong et al., 2023], BranchSBM extends this to branched multi-modal targets [Tang et al., 2025], CellFlow scales flow matching to phenotype modeling [Klein et al., 2025a], and CellStream learns dynamical OT-informed embeddings [Ling et al., 2025]. These methods use time well, but each forward query still requires iterative ODE/SDE or bridge integration, and each dataset is typically fit independently. We take PRESCIENT, BranchSBM, and CellFlow as our direct baselines; Chreode keeps their population-level view but amortizes the transition into one residual forward pass indexed by Δ and a, and pretrains a single backbone across all training trajectories.

Cell foundation models and concurrent world models. Large cell atlases can support transferable representations for cell annotation, integration, and gene-network reasoning [Theodoris et al., 2023, Cui et al., 2024], but neither Geneformer nor scGPT natively models a transition operator. Two concurrent cellular world models move toward the AI Virtual Cell agenda [Bunne et al., 2024]. AlphaCell combines a knowledge-rich decoder with OT-CFM and targets perturbation response in unseen contexts [Chuai et al., 2026]. X-Cell scales to 4.9B parameters and fits a diffusion language model on 25.6M perturbed transcriptomes [Wang et al., 2026]. Both retain multi-step integration (OT-CFM or iterative diffusion denoising) and neither imposes an explicit Waddington-style biological prior on the transition operator. Because AlphaCell and X-Cell are closed-source and trained on proprietary perturbation compendia, we do not attempt a head-to-head performance comparison; instead we position Chreode through its structural differences, namely one-step inference and the Waddington residual decomposition, and we benchmark against open task-specific systems (BranchSBM, PRESCIENT, CellFlow, CellOT, scGen).

Waddington landscapes, flux, and one-step generators. Waddington's landscape remains the core abstraction for differentiation: cells move through basins of fate rather than arbitrary Euclidean space [Waddington, 1957]. Quantitative landscape theory adds that biological paths are shaped by both potential gradients and non-equilibrium flux, not by steepest descent alone [Wang et al., 2011]. In parallel, world models in reinforcement learning learn latent state-action transitions for prediction and planning [Ha and Schmidhuber, 2018, Hafner et al., 2020]. We use the same abstraction for cells: a cytokine or a gene knockout is an action on a state, not merely metadata. Drifting Models [Deng et al., 2026] recently show that a population-matching objective can train a generator whose pushforward is evaluated in one call rather than through many integration steps. Chreode combines this one-step amortization with cell-specific state-action semantics and a Waddington-style residual prior that is, to our knowledge, the first such decomposition in a cell foundation model.

Among the open task-specific systems we benchmark against in §5, Chreode is the only entry that combines a single-step inference path, a pretrained backbone shared across downstream datasets, time and action conditioning, and a structured residual prior; a full per-method comparison on inference and structural axes is in Appendix J.

3 Method

Chreode learns a controlled population transition $p(z_{t+\Delta} \mid z_t, \mathrm{do}(a))$ from snapshot data. Figure 2 summarizes the model. Panel (a) shows the two-stage training pipeline (Stage 1 latent encoder, Stage 2 W-DiT dynamics) and the downstream transfer modes; panel (b) shows the W-DiT residual decomposition into a potential-gradient term, an antisymmetric flow term, and a stochastic-spread term, all computed in a single forward pass; the bottom row shows the population-level training objectives that align predicted and target distributions. We first formalize the one-step prediction setup (§3.1), then motivate the three-component biological residual decomposition (§3.2), instantiate it as a DiT with decoupled time embeddings for the gradient and rotational branches (§3.3), and train it with a population-matching loss that operates on unpaired snapshots (§3.4). The two design choices that distinguish Chreode from an unconstrained DiT residual head are the antisymmetric flow component and the decoupled time codes; we ablate both in §6.

3.1 Problem setup and notation

Let $x \in \mathbb{R}^G$ denote a cell's gene expression vector over a fixed ortholog vocabulary of G genes. A Stage 1 encoder $\mathcal{E}_\phi : x \mapsto z \in \mathbb{R}^d$ maps x to a d-dimensional latent state with $d = 128$ as the default ($d = 64$ is used in the small-scale component validation of Appendix D); a decoder $\mathcal{D}_\phi : z \mapsto x$ is available for gene-space evaluation. Stage 2 learns a one-step latent transition: given a source state

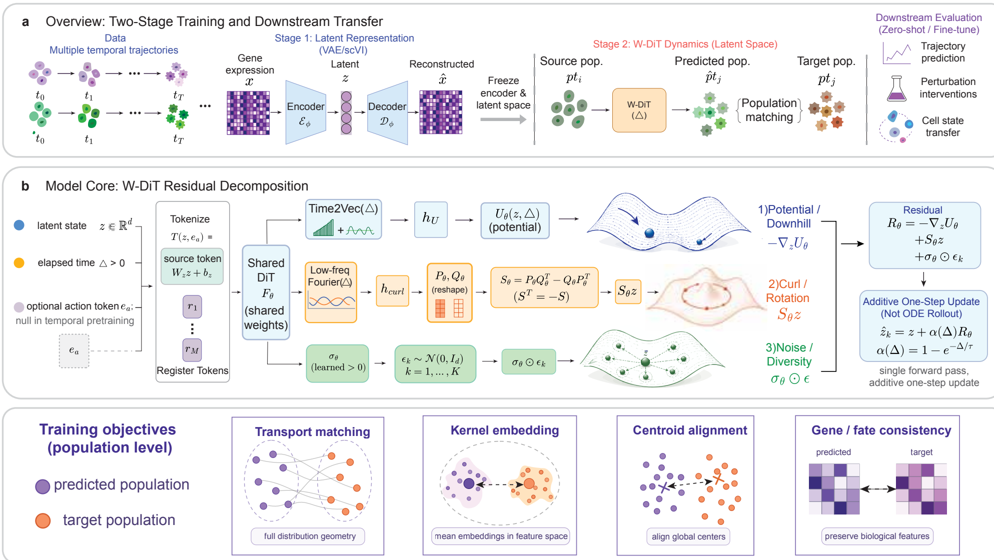

Figure 2: Chreode overview. **(a)** Two-stage pipeline: a Stage 1 scVI encoder $\mathcal{E}_\phi$ produces a frozen latent z, on top of which the Stage 2 Waddington-DiT learns a one-step latent transition $p_{t_i} \to \hat{p}_{t_j}$ matched against the observed p_{t_j}; the frozen backbone supports zero-shot and fine-tuned downstream transfer. **(b)** W-DiT residual decomposition: the state z, elapsed time Δ, and optional action token e_a are processed by a shared DiT, whose features feed three heads producing the potential gradient $-\nabla_z U_\theta$ (Time2Vec time code), the antisymmetric flow $S_\theta z$ with $S_\theta = P_\theta Q_\theta^\top - Q_\theta P_\theta^\top$ (bounded low-frequency Fourier time code), and the diagonal stochastic spread $\sigma_\theta \odot \epsilon_k$; the residual R_θ enters the one-step prediction $\hat{z}_k = z + \alpha(\Delta) R_\theta$, evaluated in a single forward pass per query. **(Bottom row)** Population-level training objectives jointly composed in Eq. (5).

$z \in \mathbb{R}^d$, an elapsed time $\Delta \geq 0$, an action a, and independent noise draws $\epsilon_1, \dots, \epsilon_K \in \mathbb{R}^d$, the model produces K stochastic samples

$$\hat{z}_k \;=\; z \;+\; \alpha(\Delta)\, R_\theta(z, \Delta, a, \epsilon_k)\,, \qquad \alpha(\Delta) = 1 - \exp(-\Delta/\tau), \qquad \tau > 0, \tag{2}$$

that together approximate the conditional population $p(z_{t+\Delta} \mid z_t = z, \mathrm{do}(a))$. The gate $\alpha(\Delta)$ enforces $\alpha(0) = 0$ so that $\Delta = 0$ collapses to the identity; τ is learnable and initialized from the median training Δ. All of $U_\theta, S_\theta, \sigma_\theta$ depend on (z, Δ, a) and are the three components of the residual R_θ.

3.2 Design principles from Waddington's landscape

Cell fate has been described for seventy years through Waddington's landscape metaphor [Waddington, 1957]: a cell rolls downhill on a developmental potential, while biological noise and rotational currents (cell-cycle progression, cyclic signaling) introduce directions that are not gradients of any scalar [Wang et al., 2011]. This motivates parameterizing the residual as three interpretable biological effects rather than an unconstrained vector field:

$$R_\theta(z, \Delta, a, \epsilon) \;=\; \underbrace{-\nabla_z U_\theta(z, \Delta, a)}_{\text{downhill potential}} \;+\; \underbrace{S_\theta(z, \Delta, a)\, z}_{\text{antisymmetric flow}} \;+\; \underbrace{\sigma_\theta(z, \Delta, a) \odot \epsilon}_{\text{stochastic spread}}\,. \tag{3}$$

Each term plays a distinct biological role: $-\nabla_z U_\theta$ drives cells toward low-potential basins of fate; the antisymmetric constraint $S_\theta^\top = -S_\theta$ makes $S_\theta z$ an in-tangent rotation rather than a gradient field, the simplest parameterization of the rotational current identified by quantitative landscape theory [Wang et al., 2011]; and $\sigma_\theta \odot \epsilon$ captures intrinsic biological variability. We do not claim $S_\theta z$ is a Helmholtz curl in the strict differential-geometric sense; we use "antisymmetric flow" as the architectural realization of the gradient-plus-rotation dichotomy, and §6 shows it is the part of the residual that an unconstrained DiT head cannot easily express. Gated by $\alpha(\Delta)$ the residual is continuous in Δ by construction. We denote this parameterization the *Waddington residual*.

3.3 Architecture: a decoupled potential / antisymmetric DiT

We parameterize U_θ, the low-rank factors of S_θ, and σ_θ with a shared DiT [Peebles and Xie, 2023] feature extractor, but condition the potential and antisymmetric branches on *different* time embeddings. The latent state token $W_z z + b_z$, an action token e_a used by action-conditioned variants of the model, and M learned register tokens are packed into a short sequence and processed by the shared DiT stack $F_\theta(\cdot, E)$, where E is a time embedding injected through adaLN-zero conditioning, yielding source-slot features $h_U(z, a, \Delta)$ and $h_{\text{curl}}(z, a, \Delta)$ for the two branches.

Why decouple the time embeddings. Long-Δ extrapolation is unstable when both branches share an unbounded Fourier code: $S_\theta z$ is norm-sensitive (unlike a gradient that integrates a scalar field), so high-frequency phases push the rotation through unseen angles at out-of-range Δ. We therefore give the potential branch a flexible learnable Time2Vec code [Kazemi et al., 2019], while restricting the antisymmetric branch to a bounded low-frequency periodic bank (periods $4, 8, 16, 32, 64, 128$) so its rotation rates always interpolate training-seen frequencies. Functional forms are in Appendix A; the effect of decoupling is isolated in §6.

Potential, antisymmetric, and noise heads. The potential is a scalar head $U_\theta = w_U^\top h_U + b_U$, and the downhill drift $-\nabla_z U_\theta$ is computed through autodiff. The antisymmetric head reshapes a linear projection of h_{curl} into two factors $P_\theta, Q_\theta \in \mathbb{R}^{d\times r}$ and forms

$$S_\theta(z, a, \Delta) \;=\; P_\theta Q_\theta^\top \;-\; Q_\theta P_\theta^\top, \qquad S_\theta^\top = -S_\theta, \tag{4}$$

so antisymmetry is exact by construction for any (P_θ, Q_θ) output and the rank of S_θ is at most $2r$. The noise scale is elementwise positive, $\sigma_\theta = \text{softplus}(\sigma_{\text{raw}}) + 10^{-4}$. Substituting back into Eq. (2) yields $\hat{z}_k = z + \alpha(\Delta)\big[-\nabla_z U_\theta + S_\theta z + \sigma_\theta \odot \epsilon_k\big]$, a one-step additive update evaluated in a single forward pass per query.

3.4 Population-level training objective

Training does not require cell-to-cell pairs, which is essential because scRNA-seq is destructive and the same cell is never observed at two time points. For a transition (t_i, t_j) and action a, we sample source cells $z \sim p_{t_i}$ and target cells $z' \sim p_{t_j}$ *independently*, generate $\{\hat{z}_k\}_{k=1}^K$, and match the generated population to the target population. The loss combines kernel MMD, entropic Sinkhorn W_2, a stop-gradient drifting-field term adapted from Deng et al. [2026], and a downhill regularizer on U_θ:

$$\mathcal{L} \;=\; \lambda_{\text{mmd}}\,\mathcal{L}_{\text{mmd}} \;+\; \lambda_{W_2}\,\mathcal{L}_{W_2} \;+\; \lambda_{\text{drift}}\,\mathcal{L}_{\text{drift}} \;+\; \lambda_{\text{down}}\,\mathcal{L}_{\text{down}}. \tag{5}$$

MMD and W_2 provide two complementary distributional matching signals; $\mathcal{L}_{\text{drift}}$ enforces that the pushforward of p_{t_i} under the residual matches p_{t_j} at equilibrium (the anti-symmetry of the drift around $q = p$ is the invariant that makes this loss well behaved); and $\mathcal{L}_{\text{down}}$ penalizes U_θ from moving uphill along the deterministic prediction.

For each trajectory $(t_1, \ldots, t_T)$ we sample training transitions uniformly from all ordered pairs (t_i, t_j) with $i < j$, not only the largest (t_1, t_T), which renders $\alpha(\Delta)$ identifiable (with a single Δ the $(\tau, \|R_\theta\|)$ pair is confounded). The action token e_a is null throughout temporal pretraining; the perturbation transfer reported in §5.4 uses the pretrained representation as a downstream gene-state embedding rather than via e_a. Full optimizer / schedule / multi-Δ details and rationale are in Appendix A.

4 Pretraining

Pretraining has two stages. Stage 1 fits a shared scVI [Lopez et al., 2018] encoder–decoder over a fixed cross-species ortholog gene vocabulary; Stage 2 freezes the encoder and fits the Waddington-DiT dynamics backbone of §3 on population transitions across the training trajectories. Held-out downstream datasets are never seen in either stage.

Data. The pretraining corpus is a 2,477,217-cell mouse embryonic atlas aggregated from seven publicly available datasets and ten leaf trajectories (tome-mouse, GSE140802, GSE115943, E-MTAB-6967, GSE106340, GSE132188, GSE275562), spanning developmental times from 0 to 19 days post-fertilization across 88 sampled timepoints; the per-dataset cell counts and time ranges are in Appendix B. Gene counts are harmonized to a single $\sim$33k-gene vocabulary and then restricted to 16,520 mouse–human one-to-one orthologs to enable cross-species transfer to human downstream datasets under the same input dimension. Expression is normalized with $\mathrm{normalize\_total}(10^4)$ followed by $\log 1p$.

Stage 1: shared scVI encoder. We train a single scVI model with latent dimension $d = 128$ (a $d = 64$ variant is reported in Appendix D), hidden 512, three layers, and a normal likelihood over $\log 1p$-normalized expression. We treat `leaf_dataset` as the technical batch covariate so that scVI removes inter-dataset and inter-lab effects, and we deliberately exclude developmental time from the batch covariate set: temporal variation is the biological signal that Stage 2 has to model and must therefore remain in the latent. The resulting encoder $\mathcal{E}_\phi$ and decoder $\mathcal{D}_\phi$ are frozen before Stage 2.

Stage 2: Waddington-DiT dynamics backbone. With Stage 1 frozen, we cache the latent cell matrix and train the dynamics backbone of §3.3 on population transitions $p_{t_i} \to p_{t_j}$ sampled uniformly from all ordered timepoint pairs within each leaf trajectory. We use AdamW ($\beta = (0.9, 0.95)$, weight decay 0.01), warmup cosine scheduling (5% warmup), batch size 512, $K = 8$ stochastic samples per source, and the loss of (5); the action token e_a is null throughout pretraining. The default model is the Small DiT (384 hidden, depth 12, heads 6, 4 register tokens); we also report a Tiny ablation (256 / 6 / 4 / 4) in Appendix B.

5 Downstream evaluation

We evaluate Chreode in two transfer modes, each on a distinct family of downstream tasks. As a *fine-tuning initialization* (§5.1, §5.2), the pretrained Stage 2 dynamics backbone is used to initialize a downstream W-DiT trained on the held-out target dataset. As a *gene-state embedding* (§5.4), the pretrained representation is injected into the existing GEARS perturbation predictor without modifying the GEARS training procedure. §5.3 reports an additional zero-shot fate-prediction evaluation that uses the frozen pretrained backbone with no downstream training.

Protocol. For time-transition tasks, source cells are encoded into the frozen scVI-128 latent, predictions are generated with the dynamics backbone at the required Δ, and metrics are computed in the shared latent after train-only standardization $\tilde{z} = (z - \mu_{\mathrm{train}})/(\sigma_{\mathrm{train}} + 10^{-6})$ applied identically to source, target, and predicted populations. All baselines are trained on the same shared scVI-128 representation with the same train/test split, so comparisons are independent of representation choice. Mean and standard deviation are reported over 3 random seeds. Per-target Veres curves and decoded gene-space metrics are in Appendix C.

5.1 Weinreb hematopoiesis

Task. Predict held-out target populations at days 4 ($d4$) and 6 ($d6$) from a day-2 source on the Weinreb in-vitro lineage-tracing dataset [Yeo et al., 2021]. The metric is Sinkhorn W_2 in the shared scVI-128 latent. Baselines are PI-SDE [Jiang and Wan, 2024], PRESCIENT [Yeo et al., 2021], a matched-architecture scratch W-DiT (same backbone trained from random initialization on Weinreb only), and two sanity baselines (identity / source replay; linear time-delta).

Table 1: Weinreb hematopoiesis. Lower is better. All rows are evaluated in the same shared scVI-128 latent with identical train/test splits and Sinkhorn W_2 implementation; mean $\pm$ std over 3 seeds where applicable.

Method	W_2 at $d4$ $\downarrow$	W_2 at $d6$ $\downarrow$
Chreode (fine-tuned)	**1.5133 $\pm$ 0.0757**	**1.6884 $\pm$ 0.0362**
Scratch W-DiT (matched)	1.6387 $\pm$ 0.0106	2.0478 $\pm$ 0.0719
PI-SDE	1.7452 $\pm$ 0.0219	1.8401 $\pm$ 0.0466
PRESCIENT	1.9096 $\pm$ 0.0124	1.8846 $\pm$ 0.0282
Identity / source replay	2.6949	4.4553
Linear time-delta	2.5326	3.7934

The pretrained backbone, used as fine-tuning initialization, achieves the lowest W_2 at both $d4$ and $d6$. The improvement over matched scratch holds across all three random seeds for both targets, isolating the value of pretraining over architecture and training budget alone. PI-SDE and PRESCIENT are trained on the same shared scVI-128 representation with the same split, so the head-to-head comparison is independent of representation choice.

5.2 Veres islet differentiation

Task. Predict each of seven target timepoints ($t = 1, \dots, 7$) from a $t = 0$ source on the Veres islet differentiation dataset [Tang et al., 2025]. We report the average per-target Sinkhorn W_2 over $t1$ through $t7$ and the $t7$ endpoint metric in the main table; the full per-target curve is in Appendix C.

Table 2: Veres islet differentiation. Lower is better. The shared scVI-128 latent and the same train/test split are used for all rows.

Method	avg W_2 over $t = 1, \dots, 7 \downarrow$	W_2 at $t = 7 \downarrow$
Chreode (fine-tuned)	**2.6171**	**2.9132 ± 0.1704**
Scratch W-DiT (matched)	2.8033	3.0892 ± 0.1144
PI-SDE	2.8296	2.9490 ± 0.0980
PRESCIENT	3.3219	3.8556 ± 0.4083
Identity / source replay	6.9355	8.2324
Linear time-delta	5.6000	6.2990

The pretrained backbone is the best row on every Veres target timepoint, including the $t7$ endpoint. The advantage holds at every $t \in \{1, \dots, 7\}$ relative to PI-SDE, PRESCIENT, and matched scratch (full curve in Appendix C).

5.3 Weinreb clonal fate prediction

Task. Predict clonal lineage outcome on the Weinreb dataset following the PRESCIENT clonal evaluation protocol [Yeo et al., 2021]: simulate stochastic endpoint trajectories from each $d2$ source cell, classify each predicted endpoint as Neutrophil/Monocyte/Other via a 20-NN classifier on $d6$ atlas cells, and score per-source fate as the predicted $\mathrm{Neu}/(\mathrm{Neu}+\mathrm{Mono})$ ratio. The metric is Pearson correlation r_{masked} between predicted and ground-truth fate ratios across clonally heldout cells. We use the frozen pretrained backbone in zero-shot mode (no Weinreb fine-tuning).

Table 3: Weinreb clonal fate prediction. Higher is better. r_{masked} is Pearson correlation on cells with at least one classified endpoint prediction; $n_{\mathrm{with\_pred}}$ is the support.

Method	$r_{\mathrm{masked}} \uparrow$	$n_{\mathrm{with\_pred}}$
Chreode (zero-shot)	**0.4680**	183
scDiffEq [Vinyard et al., 2025]	0.4627	296
moscot [Klein et al., 2025b]	0.4624	265
WOT [Schiebinger et al., 2019]	0.4593	265
Static-DiT control (zero-shot)	0.3825	335

The pretrained dynamics backbone in zero-shot mode produces fate scores competitive with strong dynamic-OT baselines (moscot, WOT) and the neural-SDE baseline scDiffEq. Replacing the dynamics-pretrained Stage 2 with a static-DiT control trained on reconstruction only drops the fate score by 0.085, isolating that the temporal dynamics objective during pretraining is what produces this transfer.

5.4 Norman Perturb-seq via GEARS embedding replacement

Task and setup. Predict treated cell populations under CRISPRi single-gene knockouts from the same-cell control population on Norman Perturb-seq [Norman et al., 2019] using the official GEARS predictor [Roohani et al., 2024]; our hypothesis is that developmental-atlas dynamics primitives (cells traversing differentiation manifolds) also describe CRISPR-induced state shifts. We test this by replacing the gene-state hidden embedding inside an otherwise-unchanged GEARS predictor with the corresponding Chreode representation. Four arms are compared under identical training (20 epochs, Norman split): the official GEARS embedding; raw scVI Stage 1 (VAE replace); the static-DiT (reconstruction only); and the dynamics-pretrained Stage 2 (Dynamics-DiT replace). We report the shared-vocabulary DE20 metric (top-20 differentially expressed genes selected within the gene set shared by our ortholog vocabulary and Norman).

Table 4: Norman Perturb-seq. GEARS gene-state embedding replaced with three Chreode arms and an scVI baseline; identical GEARS training. Lower DE20 MSE / higher $r, \Delta r$ are better.

Run	DE20 MSE ↓	DE20 r ↑	DE20 Δr ↑
GEARS official	0.21208	0.79911	0.75931
GEARS + scVI replace	0.21262	0.79231	0.75556
GEARS + Static-DiT replace	0.19358	0.80789	0.77007
GEARS + **Dynamics-DiT** replace	**0.18580**	**0.81288**	**0.77135**

The dynamics-pretrained representation reduces DE20 MSE by 12.4% relative to unmodified GEARS. The progression scVI (neutral) → Static-DiT (-8.7%) → Dynamics-DiT (-12.4%) isolates the temporal-pretraining contribution: the Static-DiT → Dynamics-DiT step comes specifically from the dynamics objective encoding how states evolve along developmental trajectories. A velocity-consistency evaluation in CellStream-defined latents (Appendix H) and a per-query timing comparison (Appendix G) provide complementary evidence.

6 Ablations

We ablate the architectural and training choices of §3 by re-pretraining Stage 2 with one component swapped per row, then fine-tuning each pretrained checkpoint on Weinreb and Veres under the same downstream protocol as Tab. 1/Tab. 2 (3 seeds, 5000 epochs, shared scVI-128 latent). *Group 1 (architecture)* tests whether the Waddington residual itself is the source of gain: **(a)** replacing (3) with an unconstrained DiT head $R_\theta = \mathrm{DiT}_\theta(z, \Delta, a, \epsilon) \in \mathbb{R}^d$ (no potential / antisymmetric / noise factorization); **(b)** tying both branches to the same Time2Vec embedding; **(c)** tying both to the bounded low-frequency Fourier code. *Group 2 (training recipe)* holds the architecture fixed and tests whether the population objective and multi-Δ schedule are load-bearing: **(d)** single-Δ training (endpoint (t_1, t_T) only); **(e)** dropping $\mathcal{L}_{\mathrm{drift}}$; **(f)** dropping $\mathcal{L}_{\mathrm{down}}$. Chreode is best on the benchmark-level averages (Weinreb avg and Veres avg, average rank 1.0); single-Δ training is the most damaging swap, consistent with all-ordered multi-Δ training being necessary for identifying $\alpha(\Delta)$ at intermediate horizons. Full numbers (Table 10) and per-target details are in Appendix K; small-scale component validation is in Appendix D.

7 Scope and Future Work

Chreode pretrains a single backbone whose representation supports two distinct transfer modes (fine-tuning initialization for time-transition prediction; gene-state embedding for perturbation prediction inside GEARS). The Waddington residual decomposition makes one-forward-pass training feasible under a population-matching objective, and the developmental-atlas recipe makes the representation reusable across the time-transition / perturbation-prediction divide.

Why developmental dynamics transfers to perturbation prediction. The Norman improvement (§5.4) is, to our knowledge, the first empirical evidence that a dynamics representation pretrained purely on developmental atlases provides a trajectory prior for genetic perturbations on an unrelated dataset; we interpret this as both differentiation and CRISPR-induced shifts traversing cell-state manifolds along coherent trajectories, suggesting that scaling developmental-trajectory pretraining is a productive direction for foundation-scale perturbation prediction.

Scope and future work. Cross-species transfer relies on the 16,520-gene ortholog vocabulary; adult human tissues are not evaluated. Time-transition transfer uses fine-tuning rather than strict zero-shot, the perturbation arm reuses GEARS, and pretraining scale is one to two orders below large world models on proprietary compendia. Future work (Appendix I): scaling the corpus, a fate-aware downstream criterion, and a gene-aware perturbation encoder end-to-end with the backbone.

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Setting	Value
Gene vocabulary	16,520 mouse–human 1:1 orthologs (Appendix E)
Expression transform	`normalize_total`(10^4) then $\log 1p$
Likelihood	Normal
Latent dimension d	128 (default); 64 used in small-scale component validation
Hidden dimension	512
Depth	3 encoder / 3 decoder layers
Batch covariate	`leaf_dataset` (10 leaves)
Timepoint batch correction	disabled (temporal signal retained for Stage 2)
Optimizer	Adam (scvi-tools default)
Batch size	4096
Epochs	2 ($\approx$ 4.95M cell visits, validation plateaued)

Setting	Value
Backbone size	DiT Small: hidden 384, depth 12, heads 6, 4 register tokens
Curl low-rank dim r	16
U time embedding	Time2Vec, $m = 8$ periodic channels, bounded $\omega_{\max} = \pi$
Curl time embedding	Bounded low-frequency Fourier, periods $(4, 8, 16, 32, 64, 128)$
Δ normalization	$\Delta_{\text{scale}} = \tau_{\text{init}} \cdot \ln 2$, per leaf trajectory
Gate $\alpha(\Delta)$	$1 - \exp(-\Delta/\tau)$, τ learnable
σ_θ	$\text{softplus}(\sigma_{\text{raw}}) + 10^{-4}$, elementwise
Stochastic samples K	8 (pretraining), 32 (downstream evaluation)
Loss	MMD + Sinkhorn W_2 + drifting field + downhill regularizer
Loss weights	λ_{mmd}=1, λ_{W_2}=1, λ_{drift}=1, λ_{down}=0.1
Multi-Δ sampling	uniform over all ordered (t_i, t_j) pairs per leaf
Optimizer	AdamW, $\beta = (0.9, 0.95)$, weight decay 0.01
Learning-rate schedule	warmup cosine, 5% warmup fraction
Base learning rate	3×10^{-4}
Gradient clip	1.0
Batch size	512 source cells per step
Training steps	3,356
EMA	disabled (found not beneficial in component validation)
Seed	single seed; downstream evaluation uses 3 seeds of data/noise draws

A Hyperparameters

A.1 Stage 1: shared scVI encoder

A.2 Stage 2: Waddington-DiT dynamics backbone

Multi-Δ training rationale. For each leaf trajectory with timepoints $(t_1, \ldots, t_T)$, training transitions are sampled uniformly from all ordered pairs (t_i, t_j) with $i < j$, not only from the largest (t_1, t_T). This is what renders $\alpha(\Delta)$ identifiable from data: with a single observed Δ, the pair $(\tau, \|R_\theta\|)$ is confounded because increasing τ and rescaling R_θ produces the same one-step prediction; only with multiple Δ's does the shape of $\alpha(\Delta)$ become uniquely pinned. The action token e_a is null throughout pretraining on mouse embryonic atlas trajectories, so the model learns action-free temporal dynamics; the perturbation transfer reported in §5.4 uses the pretrained dynamics representation as a downstream gene-state embedding rather than via e_a. The full ablation against single-Δ training is in Appendix K.

A.3 Compute resources

Stage 1 scVI training uses 1 GPU. Stage 2 dynamics pretraining uses 1 GPU. Each downstream evaluation run uses a single GPU.

A.4 Downstream evaluation

For each downstream task, the frozen encoder maps source cells to the scVI latent, and the frozen dynamics backbone produces $K = 32$ stochastic samples per source cell at the required (Δ, a). We report mean and standard deviation across 3 random seeds that control data splits and evaluation noise draws; the pretrained backbone itself is a single checkpoint.

Metric definitions. Sinkhorn W_2 uses entropic regularization $\varepsilon = 0.1$ and 100 Sinkhorn iterations. MMD uses a sum of RBF kernels over bandwidths $\{0.001, 0.01, 0.1, 1, 10, 100\}$. Fate Pearson r_{masked} on Weinreb uses the clonal lineage ground truth reconstructed from publicly released Klein-lab annotations, evaluated only on clones with at least two daughter cells. The shared-vocabulary DE20 metric on Norman is computed by selecting the top-20 differentially expressed genes per condition within the gene set shared by our pretraining ortholog vocabulary and the Norman vocabulary, then computing per-condition mean squared error and Pearson correlation between predicted and observed log fold changes on those 20 genes.

B Pretraining details

Pretraining corpus breakdown. Table 5 reports the per-dataset cell counts and time ranges of the 2,477,217-cell pretraining corpus referenced in §4. The seven datasets cover ten leaf trajectories spanning developmental times from 0 to 19 days post-fertilization across 88 sampled timepoints.

Table 5: Pretraining corpus. 2.48M cells, seven datasets, ten leaf trajectories, 16.5k ortholog genes.

Dataset family	h5ad files	Cells	Time range (days)
tome-mouse	19	1,658,968	3.5 – 13.5
GSE140802 (3 leaves: inVitro, inVivo, cytokine)	9	378,135	2.0 – 16.0
GSE115943 (C1 + C2)	80	181,188	0.0 – 19.0
E-MTAB-6967	9	130,006	6.5 – 8.5
GSE106340	11	68,339	0.0 – 17.0
GSE132188	4	37,977	12.5 – 15.5
GSE275562	3	22,604	14.5 – 16.5
Total	**135**	**2,477,217**	

Ortholog vocabulary. The cross-species ortholog table is built from Ensembl BioMart, filtered to mouse–human one-to-one orthologs with `ortholog_confidence`$= 1$. This yields $16{,}520$ gene pairs. The mouse vocabulary of the pretraining atlas ($\sim$33k genes) is intersected with this table, yielding $16{,}485$ genes present in both; the remaining 35 genes are mapped via symbol aliases.

Stage 1 reconstruction quality. Stage 1 is trained for 2 epochs (1,678 optimizer steps, ≈ 4.95M cell visits) until the validation objective plateaued. The selected checkpoint reaches a held-out reconstruction Pearson of 0.586, validation negative ELBO of 0.774, latent mean -0.017, and latent standard deviation 0.84.

Stage 2 training curves. After 3,356 training steps, the last-10-batch median of the loss components is MMD 0.037, Sinkhorn W_2 31.5, drift 8.68, and downhill ≈ 0, decreasing monotonically from first-10 medians of $0.21/81.2/8.83/\approx 0$ respectively. The held-out temporal evaluation reaches Sinkhorn $W_2 = 37.5$, MMD $= 0.052$, and Δ Pearson $= 0.551$. No NaN or divergence was observed.

C Downstream evaluation details

Weinreb. Per-timepoint W_2 at $d4$ and $d6$ for the Chreode (fine-tuned), scratch, PI-SDE, PRESCIENT, and sanity-baseline comparison, plus fate Pearson breakdowns and per-clonal-size fate predictions for the zero-shot fate evaluation.

Veres. Per-timepoint W_2 across $t = 1, \ldots, 7$ for the Chreode (fine-tuned), scratch, PI-SDE, and PRESCIENT comparison.

Norman. Per-condition DE20 mean squared error and Pearson correlation breakdowns for the four GEARS arms (official, VAE replace, Static-DiT replace, Dynamics-DiT replace), and the subset of conditions whose orthologous gene coverage exceeds 80%.

D Small-scale component validation

Before pretraining at scale, we validated the Waddington-DiT residual architecture and the training recipe on small-scale from-scratch benchmarks on Weinreb ($d2 \to d6$) and Veres (7 timepoints). These numbers do not count as foundation-model contributions and are reported here as engineering evidence that each architectural decision (decoupled time embeddings, additive vs Cayley update, multi-Δ training, loss balancer) produces the expected effect at small scale.

Selected architectural decisions validated at small scale. (i) Multi-Δ training improves intermediate-timepoint Weinreb W_2 over single-Δ training by a large margin. (ii) Bounded low-frequency Fourier time features in the antisymmetric branch produce stable long-Δ behavior, whereas tying both branches to a shared unbounded Fourier code is unstable beyond the training horizon. (iii) Using Time2Vec in the potential branch while keeping bounded low-frequency Fourier in the antisymmetric branch gives the best tradeoff between in-distribution accuracy and long-Δ stability. Full ablation tables are available in the anonymized repository.

E Licenses for existing assets

Table 6: Pretraining and downstream datasets used in this paper. All datasets are publicly available.

Dataset	Accession / source	Notes
E-MTAB-6967	ArrayExpress E-MTAB-6967	mouse gastrulation
GSE106340	GEO GSE106340	mouse embryo
GSE115943 (C1, C2)	GEO GSE115943	mouse embryo, two replicates
GSE132188	GEO GSE132188	mouse embryo
GSE140802	GEO GSE140802	inVitro / inVivo / cytokine leaves
GSE275562	GEO GSE275562	mouse embryo
tome-mouse	Qiu et al. [2022] data release	mouse development atlas
Weinreb hematopoiesis	GEO GSE140802 subset	as released with Yeo et al. [2021]; used for both time-transition and fate evaluation
Veres islet	as released with Tang et al. [2025]	7-timepoint differentiation
Norman Perturb-seq	GEO GSE133344	[Norman et al., 2019]
CellStream EMT, MOSTA	shipped with Ling et al. [2025]	appendix velocity-consistency evaluation

Baseline software: PRESCIENT [Yeo et al., 2021] under its GitHub license; BranchSBM [Tang et al., 2025] under its GitHub license; CellFlow [Klein et al., 2025a] under its release license; CellOT [Bunne et al., 2023] and scGen [Lotfollahi et al., 2019] under their respective open-source licenses. The scVI framework [Lopez et al., 2018] is used under the scvi-tools BSD-3 license.

F Broader impacts

Positive impacts. A cell world model that predicts controlled state transitions in one forward pass can reduce the cost and latency of in-silico screens for drug and genetic perturbations, allowing wet-lab experiments to be more targeted and accelerating the prioritization of therapeutic candidates. By reusing a single pretrained backbone across developmental and perturbational downstreams, Chreode also consolidates a landscape of previously task-specific systems, which makes reproducibility, auditability, and iterative refinement easier for the community.

Potential negative impacts. Any model capable of predicting cellular responses to perturbations shares a generic dual-use concern: a sufficiently accurate predictor could in principle be used to nominate interventions with harmful biological effects. We note that Chreode is trained on developmental transcriptomic atlases rather than on pathogen, toxin, or human-clinical data, so the kinds of interventions it is exposed to at pretraining time are developmental signals rather than cytotoxic or infection-related perturbations. Readers interested in dual-use policy for perturbation-prediction models are referred to ongoing community discussions around responsible release of biological foundation models.

Release. We release the Chreode codebase and raw downstream predictions under an open research license. Pretrained weights are not released with this submission.

G Inference-cost comparison

Table 7 reports the per-query inference cost of Chreode and the dominant multi-step baselines benchmarked in §5, measured on a single NVIDIA A100.

Table 7: Per-query inference cost on a single NVIDIA A100, fp32, batch size 1. "NFE" counts the number of network forward passes per (source cell, Δ, action) query. Wall-clock and FLOPs are reported as median over 60 Weinreb $d2 \to d6$ queries with 8 warmup queries discarded; FLOPs from `torch.profiler`.

Method	NFE / query	ms / query	GFLOPs / query
PRESCIENT	$\sim$88	194.33	0.07
CellFlow	50–100 ODE	410.36	—
Chreode	**1**	64.79	0.90

Chreode resolves a query in a single DiT-Small forward pass and is therefore $3.0\times$ faster than PRESCIENT and $6.3\times$ faster than CellFlow at single-query latency, even though its forward pass includes the autograd backprop required by the $-\nabla_z U_\theta$ head. The result projects directly onto screen scale: a 10^8-query screen on a single A100 takes approximately 1,800 hours for Chreode versus 5,398 hours for PRESCIENT and 11,399 hours for CellFlow. CellFlow GFLOPs are not reported because its JAX/diffrax solve is outside the PyTorch profiler.

H Velocity consistency in CellStream-defined latent spaces

As an additional check on whether the pretrained backbone produces smooth, locally-consistent velocity fields, we evaluate Chreode in the latent spaces defined by the CellStream pretrained encoder [Ling et al., 2025] on three datasets shipped with that codebase. We use Chreode (fine-tuned) as the dynamics arm, with the pretrained Stage 2 backbone fine-tuned on each dataset's CellStream latent for the same number of epochs as the local CellStream comparison. The metric is kNN-20 velocity consistency (VC) and Sinkhorn W_2 at the endpoint timepoint, both computed in CellStream's own latent space.

Table 8: Velocity consistency in CellStream-defined latent spaces. VC is kNN-20 cosine consistency of predicted local velocity ($\uparrow$); endpoint W_2 is Sinkhorn W_2 at the dataset's last observed timepoint ($\downarrow$). Mean over 3 seeds where applicable.

Dataset	Method	VC kNN20 $\uparrow$	endpoint W_2 $\downarrow$
EMT	CellStream (pretrained)	0.9743	–
EMT	**Chreode** (fine-tuned)	$\mathbf{0.9983 \pm 0.0002}$	–
MOSTA	CellStream (pretrained)	0.9638	0.0358 ± 0.0025
MOSTA	**Chreode** (fine-tuned)	$\mathbf{0.9992 \pm 0.0003}$	$\mathbf{0.0355 \pm 0.0009}$

The pretrained backbone, used as fine-tuning initialization in the CellStream-defined latent, produces velocity fields whose local consistency exceeds the CellStream pretrained model on EMT and MOSTA, and matches CellStream's MOSTA endpoint W_2. We treat this as evidence that the dynamics representation captures meaningful local direction, complementary to the in-distribution scVI-128 results in §5.

I Scope notes

Pretraining corpus is mouse-embryonic. Cross-species transfer to human downstream datasets is mediated only by the $16{,}520$-gene mouse–human ortholog vocabulary; we do not pretrain on any human transcriptomes. A richer cross-species pretraining schedule that includes adult human tissues is left to future work.

Time-transition transfer uses fine-tuning, not zero-shot. Weinreb and Veres results in §5.1, §5.2 use the pretrained backbone as initialization for downstream fine-tuning. The clonal fate evaluation in §5.3 uses the frozen pretrained backbone in zero-shot mode. Whether a single training recipe can cover both regimes simultaneously, including a clone-aware downstream criterion, is an open question.

Perturbation transfer reuses GEARS rather than a standalone operator. The Norman result in §5.4 comes from injecting the pretrained representation as the gene-state embedding inside an unmodified GEARS predictor. We do not present a standalone perturbation operator from the pretrained backbone in this paper; designing a gene-aware perturbation encoder that composes end-to-end with the dynamics backbone is a natural follow-up.

Scale is below concurrent large cellular world models. Our pretraining scale (2.4M cells, Small DiT, of order 10^7 parameters) is one to two orders of magnitude below concurrent large cellular world models such as AlphaCell [Chuai et al., 2026] and X-Cell [Wang et al., 2026], which train on proprietary perturbation compendia. Whether the present transfer modes scale to 100M-cell developmental atlases is an open empirical question.

No comparison to representation-only foundation models. We cite Geneformer and scGPT [Theodoris et al., 2023, Cui et al., 2024] but do not benchmark against them in this paper, because they do not expose a transition operator and require a nontrivial adapter head to be evaluated on our downstream regimes. Comparison to representation-only foundation models is orthogonal to the present contribution and is left to follow-up work.

Latent-space evaluation by default. W_2, MMD, and energy-distance metrics are computed in the shared scVI-128 latent; gene-space metrics for Norman use the Stage 1 decoder. Additional gene-level evaluation panels are reported in §C.

No uncertainty quantification yet. The model exposes a per-dimension noise scale $\sigma_\theta(z, \Delta, a)$, but we do not currently calibrate or report posterior credible intervals on downstream predictions. Calibration of the stochastic spread against held-out replicate variance is a natural follow-up.

J Method comparison details

Table 9 expands on the inference and structural differences between Chreode and the open task-specific systems we benchmark against in §5. We report each method along five axes: per-query inference cost, whether a pretrained backbone is shared across downstream datasets, whether the operator is conditioned on elapsed time Δ, whether it accepts an action input a, and what architectural inductive bias (if any) the residual / drift carries.

Chreode is the only entry that combines (i) a single-step inference path, (ii) a pretrained backbone shared across all downstream datasets rather than retrained per dataset, (iii) joint time and action conditioning, and (iv) a structured residual prior that decomposes the velocity field into a potential gradient, an antisymmetric flow, and a stochastic spread. Among the multi-step rows, PRESCIENT carries the closest structural inductive bias (a potential landscape) but no antisymmetric component and no shared pretraining; among the one-step rows, scGen and CPA add latent-additive priors but discard time entirely. We do not include the closed-source AlphaCell and X-Cell systems in this table because their architectural details and inference protocols are proprietary.

Table 9: Method comparison on inference and structural axes. "Inference cost" uses each method's typical released setting; "architectural residual prior" refers to architectural inductive bias rather than data preprocessing.

Method	Inference cost	Pretrained backbone	Time-cond. (Δ)	Action-cond. (a)	Architectural residual prior
PRESCIENT	$\sim$88 Euler steps	per dataset	yes	limited	potential landscape
BranchSBM	multi-step SDE	per dataset	yes	no	none
CellFlow	40–100 ODE steps	per dataset	yes	yes (control)	none
scGen	1 step (no time)	per dataset	no	yes	latent additive
CPA	1 step (no time)	per dataset	no	yes	additive composition
Chreode	**1 step**	**2.4M cells**	$\alpha(\Delta)$ **gate**	**token**	**antisymmetric flow + potential**

K Ablation details

This appendix expands on the architectural and training ablations summarized in §6. Each row of Table 10 is a separately pretrained Stage 2 checkpoint with one component swapped relative to the selected Chreode configuration; Stage 1 (the scVI encoder) is held fixed across all ablation rows. Each pretrained checkpoint is then fine-tuned on Weinreb and Veres under the same downstream protocol as Tab. 1/Tab. 2 (3 seeds, 5000 epochs, shared scVI-128 latent), so absolute Sinkhorn W_2 values are directly comparable to the main downstream tables. The headline columns are benchmark-level averages (Weinreb avg over $\{d4, d6\}$ and Veres avg over $t1$ through $t7$) plus the average rank across the two; per-target detail is provided in the last two columns.

Table 10: Architectural and training ablations under matched downstream fine-tuning. Headline columns are benchmark-level averages (Weinreb avg over $d4/d6$, Veres avg over $t1$ through $t7$) plus the average rank across both. Lower is better.

Variant	Weinreb avg $\downarrow$	Veres avg $\downarrow$	Avg rank $\downarrow$	Weinreb $d4$ $\downarrow$	Weinreb $d6$ $\downarrow$
Chreode (selected)	**1.6008**	**2.6171**	**1.0**	**1.5133**	1.6884
G1: architecture					
Unconstrained DiT residual	1.8922	3.4030	2.0	2.1329	**1.6516**
Tied time embeddings (both Time2Vec)	1.9073	3.7139	5.0	2.0868	1.7278
Tied time embeddings (both low-freq)	1.9416	3.5298	4.0	2.1254	1.7578
G2: training recipe					
Single-Δ training	2.0579	3.7938	7.0	2.1313	1.9845
Without $\mathcal{L}_{\mathrm{drift}}$	1.8974	3.6258	3.5	2.0878	1.7071
Without $\mathcal{L}_{\mathrm{down}}$	1.9835	3.6793	5.5	2.1015	1.8655

Reading the headline. The selected Chreode is the best row on both benchmark-level aggregates (Weinreb avg 1.6008, Veres avg 2.6171) and obtains the best average rank across the two. The Veres advantage is the most decisive: every ablation row exceeds Chreode's Veres average by 30–45% (3.40–3.79 vs 2.6171), which is consistent with the design target of multi-target temporal transfer.

Reading G1 (architecture). *Unconstrained DiT residual* (no potential / antisymmetric / noise factorization) is the closest competitor at the Weinreb $d6$ endpoint (1.6516 vs 1.6884, within seed variance: std 0.21 vs 0.04), but its Weinreb $d4$ (2.1329 vs 1.5133, $+41\%$) and Veres average (3.4030, $+30\%$) are substantially worse: the unstructured head can fit a single endpoint but cannot match the structured residual on early targets or multi-target Veres, isolating the value of the Waddington decomposition for multi-Δ transfer. *Tied Time2Vec* (both branches share the unbounded Time2Vec code) increases Veres average to 3.7139 ($+42\%$); we attribute this to the antisymmetric branch becoming unstable at long Δ when its rotation rates drift through unseen frequencies, consistent with the architectural rationale in §3.3. *Tied low-frequency Fourier* (both branches share the bounded code) reaches Veres average 3.5298 ($+35\%$): a single bounded code is not flexible enough to match the long monotone developmental schedule that the potential branch needs.

Reading G2 (training recipe). *Single-Δ training* is the most damaging swap in either group (Weinreb avg 2.0579, Veres avg 3.7938, average rank 7.0). This matches the design rationale that all-ordered multi-Δ training is necessary for identifying the time gate $\alpha(\Delta)$ at intermediate horizons (Appendix A, "Multi-Δ training rationale"): when only the largest Δ is observed at training time, the model has no signal to disambiguate the gate shape from the residual norm. *Without* $\mathcal{L}_{\mathrm{drift}}$ (Weinreb avg 1.8974, Veres avg 3.6258) and *without* $\mathcal{L}_{\mathrm{down}}$ (Weinreb avg 1.9835, Veres avg 3.6793) both underperform the selected Chreode by 20–40% on the aggregates; removing $\mathcal{L}_{\mathrm{down}}$ hurts Weinreb more (Weinreb $d6$ 1.8655 vs 1.6884), while removing $\mathcal{L}_{\mathrm{drift}}$ hurts Veres more, indicating dataset-dependent value rather than a uniform monotone effect, but both regularizers help on average and we keep them in the selected configuration.

Detailed Weinreb metrics, selected Chreode vs unconstrained DiT. Because Weinreb $d6$ full-latent W_2 is close between the selected configuration and the unconstrained DiT (within seed variance), Table 11 reports additional Weinreb metrics from the same fine-tuned runs: Sinkhorn W_1 and W_2 on the top-2 principal components, full-latent W_1, BranchSBM-style fixed-bandwidth MMD, and our unbiased median-heuristic MMD. At $d4$ the selected configuration is better on every reported metric. At $d6$ the selected configuration is better on top-2 W_1/W_2 and both MMD statistics, while unconstrained DiT is slightly better on full-latent W_1/W_2 but with $5\times$ larger seed variance. The combined picture supports the headline reading: the structured residual gives more stable distribution matching at $d4$ and on low-dimensional / kernel metrics at $d6$, while a single full-latent $d6$ endpoint metric is not enough to favor either side.

Table 11: Detailed Weinreb metrics for selected Chreode vs unconstrained DiT under fine-tuning, mean $\pm$ std over 3 seeds. Lower is better.

Method	Day	Top-2 $W_1\downarrow$	Top-2 $W_2\downarrow$	MMD full $\downarrow$	Full $W_1\downarrow$	Full $W_2\downarrow$
Chreode	$d4$	**0.046±0.005**	**0.079±0.008**	**0.0074±0.0003**	**1.286±0.031**	**1.513±0.076**
Unconstrained DiT	$d4$	0.102±0.009	0.147±0.018	0.0349±0.0027	1.860±0.005	2.133±0.007
Chreode	$d6$	**0.043±0.005**	**0.065±0.003**	**0.0050±0.0003**	1.529±0.041	1.688±0.036
Unconstrained DiT	$d6$	0.065±0.013	0.097±0.023	0.0129±0.0026	**1.393±0.103**	**1.652±0.210**

What this ablation does not test. We do not vary the latent dimension d, the DiT backbone size, the scVI encoder architecture, or the action encoder, because these are held fixed across all reported downstream evaluations and would require re-running every downstream comparison. Small-scale architectural decisions (additive vs Cayley update, register-token count, EMA toggle) are validated separately in Appendix D.

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